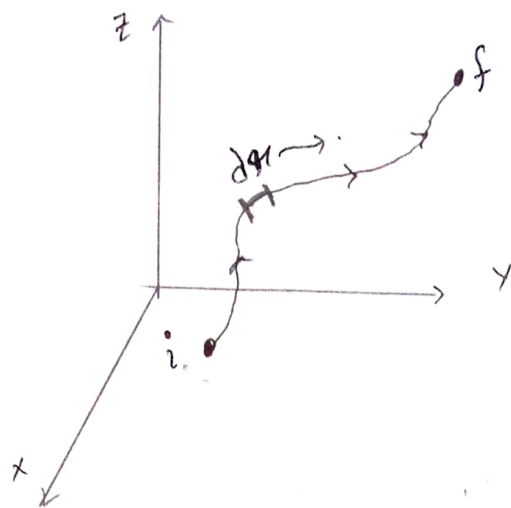


Work-Energy Theorem

①



A particle moves from initial position i to final position f

→ Define work done by particle in moving from initial position " i " to final position " f " as

$$W = \int \vec{F} \cdot d\vec{r}$$

where $d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$ is a line element along the path from i to f as shown in the figure.

$$W = \int_i^f \vec{F} \cdot d\vec{r}$$

$$= \int_i^f \left(m \frac{d\vec{v}}{dt} \right) \cdot d\vec{r}$$

$$= \int_i^f m d\vec{v} \cdot \frac{d\vec{r}}{dt}$$

$$= \int_i^f m d\vec{v} \cdot \vec{v}$$

(2)

$$\therefore W = \int_i^f m \vec{v} \cdot d\vec{v}$$

$$= m \frac{v^2}{2} \Big|_i^f$$

$$W = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W = \Delta KE$$

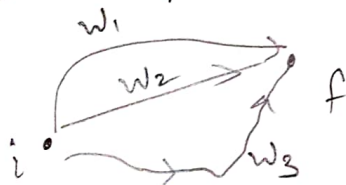
Work done in moving from initial to final position is the change in the kinetic energy of the particles.

→ This is the work-energy theorem.

Forces

Conservative

→ Work done is independent of the path taken.

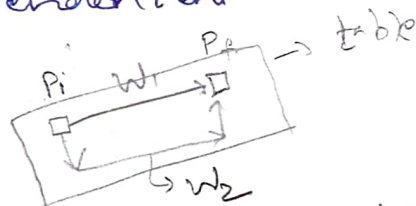


$w_1 = w_2 = w_3$ } work done is same no matter the path.

eg. gravitational force.

Non-Conservative

→ Work done is path dependent.



$w_1 \neq w_2$ for a block on table to move from i to f position.

∴ work done is path dependent

eg. frictional force.

Now define work done as

$$W = \int_i^f \vec{F} \cdot d\vec{r}$$

for conservative forces it is path dependent.

∴ we can write

$$W = \int_i^f \vec{F} \cdot d\vec{r} = -U(r_f) + U(r_i)$$

where $U(r_f)$ = Potential energy of particle at final position
&

$U(r_i)$ = Potential energy of particle at initial position.

$$\therefore W = \int_i^f \vec{F} \cdot d\vec{r} = - \underbrace{\Delta U}_{\text{change in potential energy.}}$$

$$W = \int_i^f \vec{F} \cdot d\vec{r} = \Delta KE \quad \left. \vphantom{\int_i^f \vec{F} \cdot d\vec{r}} \right\} \text{from work-energy theorem.}$$

$$KE_f - KE_i = -U(r_f) + U(r_i)$$

$$KE_f + U(r_f) = KE_i + U(r_i)$$

Total mechanical energy = Kinetic energy + Potential energy.

$$\therefore KE_f + U(x_f) = KE_i + U(x_i)$$

$KE + PE \}$ Conserved.

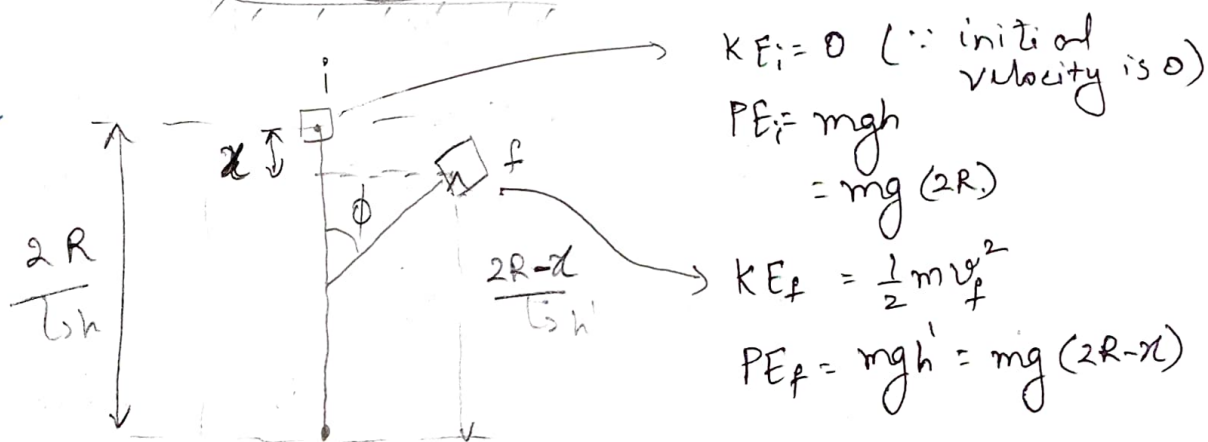
This is conservation of energy.

→ Resolve the following problem using conservation of energy.



At what ϕ will the body leave the surface?
(Assume no friction).

Sol



$$KE_i = 0 \quad (\because \text{initial velocity is } 0)$$

$$PE_i = mgh = mg(2R)$$

$$KE_f = \frac{1}{2} m v_f^2$$

$$PE_f = mgh' = mg(2R - x)$$

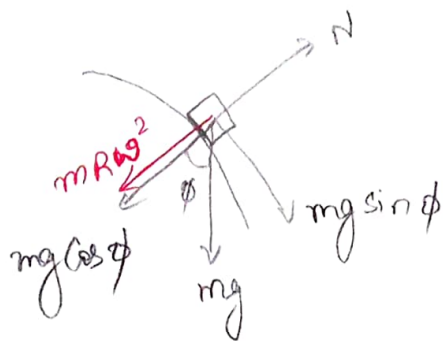
From conservation of energy ~~from~~ ⁱⁿ moving from initial to final position is

$$KE_f + U_f = KE_i + U_i$$

$$\frac{1}{2} m v_f^2 + mg(2R - x) = 0 + mg(2R)$$

$$\frac{1}{2} m v_f^2 = 2mgR - 2mgR + mgx$$

$$\underline{\underline{v_f = \sqrt{2gx}}}$$



The forces acting on the body at angle ϕ are as below.

In the ϕ direction. $mg \sin \phi$

In the x direction $N = mg \cos \phi - \underbrace{mRw^2}_{\text{centripetal force.}}$

$\therefore$ for body to leave the surface

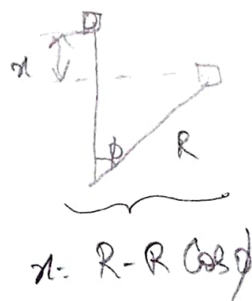
$N = 0$ & use $v = R\omega$

$$\therefore mg \cos \phi = \frac{mv^2}{R}$$

$v = \sqrt{2gx}$ from conservation of energy

$$\therefore mg \cos \phi = \frac{m \cdot 2gx}{R}$$

$$\cos \phi = \frac{2x}{R}$$



$$\cos \phi = \frac{2(R - R \cos \phi)}{R}$$

$$\cos \phi = 2 - 2 \cos \phi$$

$$\therefore \underline{\underline{\cos \phi = \frac{2}{3}}}$$

} at this ϕ the body leaves the surface.